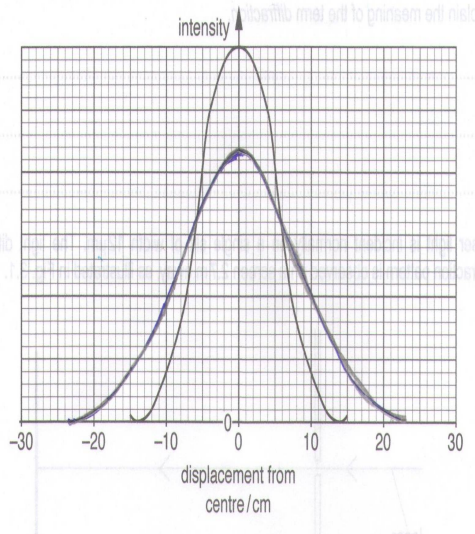


	m/s and the expression $v_{\min} = \sqrt{rg}$ is independent of mass.
3(a)	Diffraction is the bending or spreading of waves when they travel through a small opening or the spreading of waves around an obstacle. <u>Common mistake:</u> - Used the word “splitting”
3(b)	$\tan \theta = \frac{0.14}{2.7}$ $\theta = 2.968^\circ$ $\sin 2.968^\circ = \frac{\lambda}{12 \times 10^{-6}}$ $\lambda = 6.2 \times 10^{-7} \text{ m}$

3(c)	 (Wider fringe and smaller intensity at centre)
3(d)(i)	With a longer wavelength being used, the angle of diffraction for the first order minima will be larger as $\sin \theta$ is directly proportional to the wavelength of light used. This will lead to the central bright fringe to be wider.
3(d)(ii)	White light consists of different colours of light of varying wavelengths, thus will diffract at varying angles forming the coloured edges. The different colours will superpose at the central position and form the central bright fringe that is white in colour.
4(a)	Threshold frequency refers the minimum frequency of the illuminating